Sai Kiran B

Lead Software Engineer

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Senior Application Developer having **7+ years** of experience in designing, developing, and delivering robust end-to-end Windows and Web Applications. Expertise in **.NET**, **Web API**, **Entity Framework**, and **MS-SQL**, complemented by hands-on experience in frontend development technologies, including **HTML**, **CSS**, **Bootstrap** and **React**. Skilled in creating scalable, user-friendly solutions using cloud technologies like **AWS**, **Azure** and inhouse solutions on **RHEL** that meet business requirements.

PROFESSIONAL EXPERIENCE

Database and Backend Engineer
01/2025– currently

Research Assistant *08*/2023– 08/2024

North Dakota State University

- Designed and developed multiple research applications using **React**, **Python**, **.NET**, **Windows Forms**, **WPF**, **Console Applications**, **Web API** and **Shell Scripting** for delivering robust solutions for academic and research purposes.
- Created and optimized **Postgres**, **MS-SQL** database **schemas**, leveraging **SSRS** and **SSIS** for efficient reporting and integration.
- Contributed to **data visualization**, **analysis**, and insights generation for various research papers by utilizing **Python** and **R** for **advanced data analysis** and visualization techniques.
- Implemented **containerized** solutions using **Docker** and **Kubernetes**, ensuring scalable and portable application deployments for research tools.
- Developed version-controlled **research applications**, maintaining **high code quality** and collaboration and scraping of repositories using **GIT**.
- Performed application **deployment** on in-house **RHEL** infrastructure using **Podman**, **HTTPD**, and related tooling.
- Contributed to academic research by **publishing a research paper**, showcasing innovative solutions and advanced methodologies in data analysis and application design.

Programmer Analyst (SME)
03/2021– 08/2023

Fortrea *formerly called as* **LabCorp**

- Served as a Subject Matter Expert in architectural design and development of integration solutions, achieving a **45% reduction in turnaround time** for integrating Health Records and Inventory Management systems by implementing dynamically configurable solutions.
- Designed and implemented scalable integration frameworks to streamline data exchange across disparate systems using **AWS S3** and **Kafka, Azure Blob, SFTP, FTP and Tibco storage systems**.
- Enhanced system efficiency by **automating workflows**, reducing manual intervention, and minimizing errors in data synchronization using **AWS Lambda**, **AWS Glue Jobs** to manage data pipelines into **Relational** and **NoSQL Databases**.
- Collaborated with cross-functional teams to ensure seamless integration with **third-party APIs** and **legacy systems** through **Web Services** and **Web APIs** and troubleshooting using **Postman** and **Swagger**.
- Optimized the integration process by developing **configurable real-time S3 buckets**, significantly improving the turnaround time for data integration workflows.

Senior Software Engineer
07/2018– 03/2021

Calyx.ai *formerly called as* **Parexel International**

- Developed multiple clinical trial and healthcare systems utilizing **.NET MVC**, **N-Tire**, **MVVM**, **ADO.NET** and **MS-SQL**, including advanced database components such as complex **joins**, **views**, **triggers**, **stored procedures**, **functions**, **Temp Tables** and **CTEs** to support robust data management and processing.
- Built and maintained scalable Web APIs with **Swagger** and **Postman** for seamless integration and **documentation**, ensuring compatibility with various systems.
- Designed and delivered **SSRS reports**, **Crystal Reports**, and **SAS scripts** for comprehensive reporting and data analysis in clinical and healthcare environments.
- Designed scalable and secure web applications using **RBAC** and ensuring compliance of **21 CFR**, **GCP, HIPPA**, **GDPR**.
- Leveraged **GIT** and **SVN** for version control, fostering collaborative development and efficient code management.

PROJECTS

ClinPhone (Product Development – RTSM, CTMS & IRT)

- Developed and maintained **ClinPhone** systems using **.NET**, **ADO.NET**, **Entity Framework**, **HTML**, **CSS** and **Bootstrap** ensuring seamless user interfaces and enhanced functionality for clinical trial operations.
- Designed and optimized **SQL databases**, implementing **complex queries**, **Joins**, **Temp Tables**, **Functions**, **stored procedures** and **views** to manage and analyse trial data efficiently.

Pulse (Supply and Inventory Management System)

- Developed and implemented Pulse, a comprehensive Supply and Inventory Management System, utilizing **Web APIs**, **Kafka**, **AWS SQS**, **AWS S3**, and **AWS Lambda** to automate **resupply**, **stock monitoring**, and **delivery** processes.
- Designed efficient workflow **pipelines** with **AWS Step Functions** and to ensure seamless **stock tracking**, **resupply operations**, and delivery, improving accuracy, scalability, and operational efficiency across the supply chain.

Data Bridge (Integration System)

- Implemented an integration system across multiple **vendors** and **clients**, enabling efficient **data collection** and **management** from diverse systems. Utilized **AWS S3** as primary storage with **configurable S3 buckets** to adapt in real time to client standards with minimal changes, In Addition handled different systems such as **TIBCO**, **SFTP**, **FTP**, **EDC**, **Medicaid**, IBM **Market Scan**, **Optum Panther** Systems.
- Streamlined **ETL processes** to transform and load data from **XML**, **JSON**, **CSV**, and other formats into **Relational Databases** using Apache **Glue Jobs, Cloud Watch** and **SSIS tasks**, ensuring seamless and efficient data transformation and integration.

Imaging System (Proof of Concept)

- Developed a healthcare **imaging system** to integrate scans from **Azure Storages**, **Web APIs** and **React** enabling efficient collection, analysis, and anomaly detection for early identification of **tumours** and other **critical factors in patient data**.
- Utilized **AI/ML** modelling with libraries like **TensorFlow** and **OpenCV** for **pixel-level analysis** and **detection**, enhancing precision in anomaly identification. Leveraged **Python** for **data visualization** and storytelling, providing actionable insights to support early diagnosis and treatment planning.

EDUCATION

Master of Science in Computer Science & cert in Big Data Statistical Analysis, North Dakota State University (GPA: 3.7/4.0) 2024

Executive Post Graduation in Data Science, IIIT Bangalore (GPA: 3.6/4.0) 2023

Bachelor of Technology in Computer Science, Acharya Nagarjuna University (GPA: 3.6/4.0) 2018

CERTIFICATIONS

- TRec - Test Recommender Published at **ICSME** 2024.
- Certified **Azure** Fundamentals (AZ-900) by Microsoft.
- Certified Data Processing Specialist by Aspiring Minds.
- Certified **SAS** Base Programmer (SAS - 231) by SAS.

TECHNICAL SKILLS

Programming Languages:	C#, Dot Net, LINQ, ASP .Net, OOPs, Windows Forms, WPF, Ado.Net, Entity Framework, Dapper, HTML, CSS, C, XML, Json Integration. HTML, CSS, Bootstrap, React, Angular, JavaScript and TypeScript MS SQL, T-SQL, SSRS, SSIS, My SQL, Oracle, NoSQL, MongoDB, Athena, Spark. Python, SAS, R, R-Shiny.
Cloud Technologies:	AWS Dockers, Containers, Kubernetes, S3, RDS, Lambda, Glue, API Gateway Azure SQL, Azure Data Storage, Blobs, Azure Power Functions, Webservices,
Tools:	Visual Studio, VS Code, SQL Server Management Studio (SSMS) AWS Dev Kit, Azure SDK, Docker, Power Shell, Linux Shell Jupiter, Anaconda, SAS 9.4, R Studio, Tortoise SVN, Git, GitLab, Git-LFS, ServiceNow, Jira, SharePoint.
Domain Knowledge:	Supply Chain, Inventory Management, CICD, IRT, RTSM, Clinical, Health Care, Finance, HIPPA, 21 CFR, GCP, GDPR.